

CARGOMATE
SMART ROBOTIC INVENTORY SYSTEM WITH CLOUD
INTEGRATION

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Design Document (DD) of Autonomous Navigation System

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1.INTRODUCTION

1.1 Purpose

The purpose of this document is to formally describe the design decisions, system architecture, interfaces, and implementation strategies of the autonomous navigation subsystem developed as part of the warehouse robot project.

This document serves as a technical reference for evaluators, developers, and future contributors by clearly outlining how the system is structured, how components interact, and how design choices support project requirements.

1.2 Scope

This document focuses on the navigation subsystem, which is the core functional component implemented during the current development phase. Specifically, it covers:

- Navigation logic and control flow design
- Integration of multiple sensors for reliable operation
- Motor control strategy using PWM and direction control
- Risk mitigation mechanisms related to power, boot stability, and safety

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
IR	Infrared
RFID	Radio Frequency Identification
FSM	Finite State Machine
PWM	Pulse Width Modulation
ESP32	Microcontroller used for system control
GPIO	General Purpose Input/Output pins
DD	Design Document

1.4 Overview

The developed system enables a mobile robot to autonomously navigate a warehouse environment by following a predefined black-line path, identifying key locations using RFID tags, and responding safely to obstacles through ultrasonic sensing. The navigation process is fully autonomous and does not require human intervention during operation.

The system integrates multiple low-cost sensors with a finite state machine (FSM)–based control strategy implemented on an ESP32 microcontroller. Infrared sensors provide continuous path tracking, RFID ensures absolute location identification at critical points, and ultrasonic sensing acts as a safety layer by enabling immediate motor halt when obstacles are detected. This multi-sensor approach improves reliability compared to single-sensor navigation systems.

Motor control is achieved using PWM-based speed regulation and direction control through an L298N motor driver, allowing smooth and predictable movement. Careful consideration has been given to system stability, including safe startup behavior, controlled state transitions, and defined responses to all operating conditions.

2. OVERALL DESCRIPTION

The system implements an autonomous navigation subsystem for a warehouse robot using a multi-sensor approach. By integrating IR line sensors for path tracking, RFID for location identification, and ultrasonic sensing for obstacle detection, the system achieves reliable and deterministic navigation in indoor environments. The navigation logic is managed by an ESP32 microcontroller using a finite state machine, ensuring safe operation, accurate movement, and cost-effective implementation suitable for warehouse automation prototypes.

2.1 Product Perspective

Traditional low-cost mobile robots often rely on a single navigation sensor, such as an IR line sensor or ultrasonic module, which limits reliability in real-world environments.

In contrast, the proposed system integrates IR line sensors, RFID-based location identification, and ultrasonic obstacle detection, enabling more accurate navigation and decision-making.

This multi-sensor approach improves positional certainty and operational safety while remaining cost-effective.

2.1.1 System Interfaces

- ESP32 GPIO and PWM (LEDC)
- I2C interface for RFID (PN532)

2.1.2 User Interfaces

- No graphical UI at current stage
- Serial monitor used for debugging and status verification

2.1.3 Hardware Interfaces

- IR sensor array (digital inputs)
- Ultrasonic sensor (trigger/echo)
- RFID module (I2C)
- L298N motor driver (PWM + direction pins)

2.1.4 Software Interfaces

- Arduino framework for ESP32 programming
- Adafruit PN532 library for RFID communication

2.1.5 Communication Interfaces

- I2C protocol for RFID communication
- GPIO-based digital signaling for sensors and motor control

2.1.6 Memory Constraints

- ESP32 internal RAM and flash memory limitations
- Lightweight state-based logic used to minimize memory usage

2.1.7 Operations

Normal system operation follows these steps:

1. Power-on and initialization
2. Sensor stabilization delay
3. Autonomous navigation begins
4. RFID-based stop or reverse decisions
5. Obstacle detection and safe motor halt
6. Resume or terminate operation based on system state

2.1.8 Site Adaptation Requirements

CargoMate operates optimally under specific warehouse conditions:

- Black line path layout: Clearly marked black lines guide IR sensors for navigation. High contrast is essential for accurate tracking.
- RFID tags at targets and destinations: Tags at junctions and pick/drop points enable precise location identification and task execution.
- Flat indoor surface: Smooth, level floors ensure reliable line-following and stability.

- Stable power supply: Continuous power via batteries or charging stations is necessary for uninterrupted operation and cloud connectivity.

2.2 Product Functions

Key autonomous functions include:

- IR-based line following (forward/reverse): Five IR sensor array maintain alignment along black lines, with dynamic motor adjustments.
- RFID-based location and task logic: Reads RFID tags to identify locations and determine task actions via Firebase.
- Ultrasonic obstacle detection: Stops and performs avoidance maneuvers when obstacles are within ~20 cm.
- FSM-driven navigation: Finite State Machine manages movement, obstacle handling, line recovery, and task execution for predictable behavior.

2.3 User Characteristics

- Warehouse operators: Minimal interaction; mainly monitor status via cloud.
- Technicians: Basic robotics knowledge for maintenance; no programming needed.
- Autonomous operation: Fully automated navigation and inventory management without manual intervention.

2.4 Constraints

- Indoor use only: Not suitable for outdoors or uneven terrain.
- Line-dependent navigation: Cannot navigate outside predefined paths.
- Limited prototype payload: Supports light loads; future models may carry heavier items.

2.5 Assumptions and Dependencies

- Adequate lighting: Ensures IR sensor accuracy.
- Accurate RFID placement: Prevents misreads at junctions.
- Stable power supply: Required for uninterrupted operation and cloud synchronization.

2.6 Apportioning of Requirements

Phase	Focus Area	Description
Phase 1	Core Navigation	IR-based line following, RFID-based location identification, and obstacle detection with safe motor halt
Phase 2	Navigation Robustness	Line loss detection and recovery to re-acquire the path when the line is temporarily lost
Phase 3	Advanced Obstacle Handling	Obstacle avoidance logic to bypass obstacles and safely rejoin the predefined path (ongoing)

3. SPECIFIC REQUIREMENTS

3.1 External Interface Requirements

3.1.1 User Interfaces

- No direct user interface
- Fully autonomous operation

3.1.2 Hardware Interfaces

- ESP32 ↔ IR sensors
- ESP32 ↔ RFID (PN532)
- ESP32 ↔ Ultrasonic sensor
- ESP32 ↔ L298N motor driver

3.1.3 Software Interfaces

- Embedded C++ (Arduino IDE)
- PN532 driver library

3.1.4 Communication Interfaces

- I2C for RFID
- GPIO-based signaling

3.2 Architectural Design

3.2.1 High-Level Architecture Design

The high-level architecture defines a layered navigation system where sensor inputs are processed by an ESP32-based finite state controller, which generates real-time motor control commands to achieve autonomous line following, location-based stopping, and safe obstacle response.

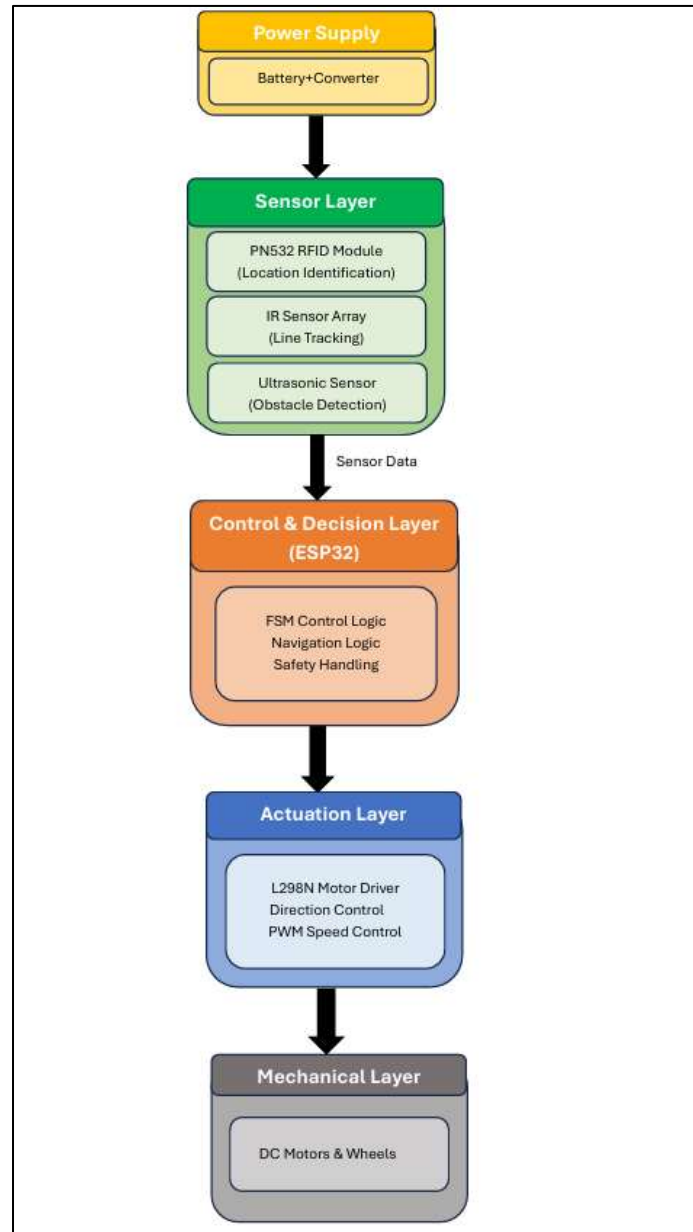


Figure 3.2. 1: High-level architecture of the ESP32-based multi-sensor autonomous navigation system

3.2.2 Process Flow Diagram

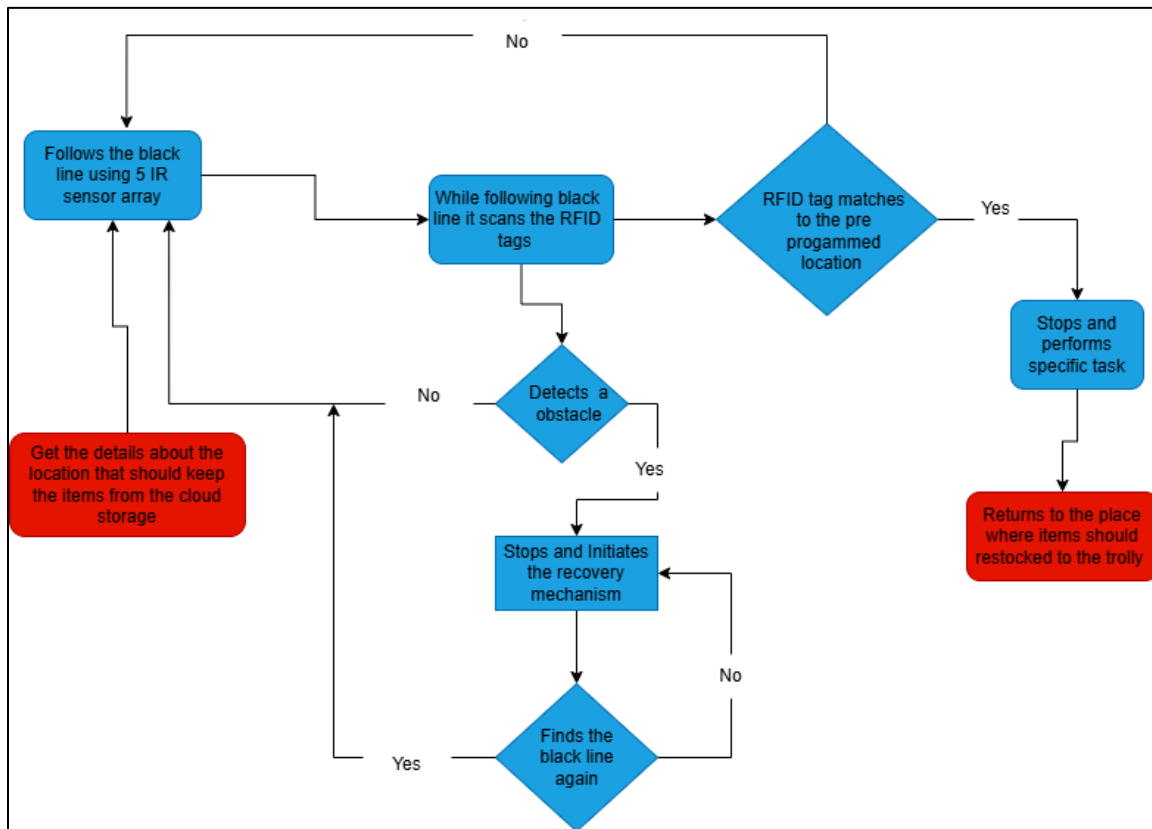


Figure 3.2. 2: Process Flow Diagram

3.2.3 Circuit Diagram

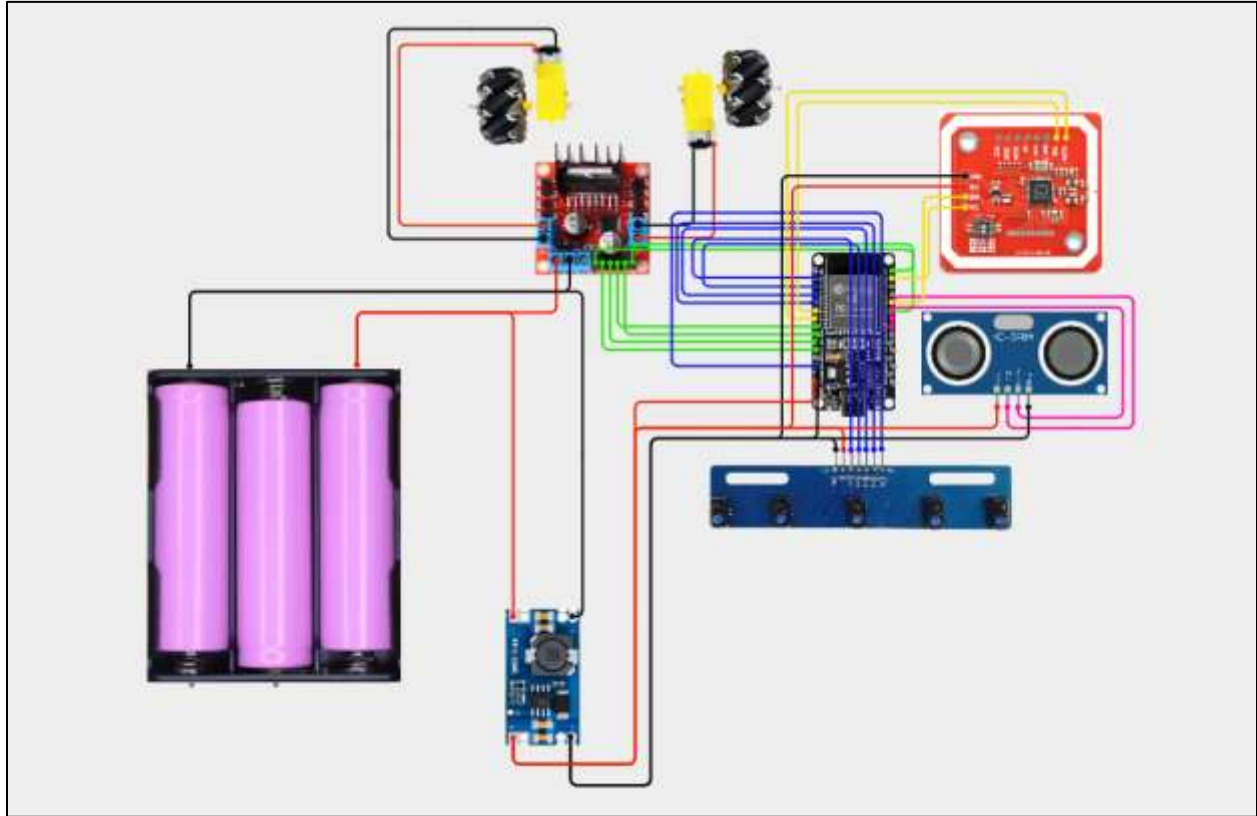


Figure 3.2. 3 : Wiring diagram of the ESP32-based navigation system

3.2.3 Hardware & Software Requirements with Justification

- ESP32 Microcontroller – Provides real-time control, FSM execution, and PWM (LEDC) motor speed control.
- 5-Channel IR Line Sensor Array – Enables accurate black-line tracking with improved reliability over single-sensor systems.
- PN532 RFID Module – Supports absolute location identification for controlled stopping and navigation decisions.
- Ultrasonic Sensor – Acts as a safety mechanism for obstacle detection and immediate motor halt.
- L298N Motor Driver – Allows bidirectional DC motor control with PWM-based speed regulation.
- 5V DC–DC Buck Converter – Supplies stable 5V power to sensors and motor driver, protecting against voltage fluctuations.

- Embedded Software (Arduino Framework) – Ensures modular, FSM-based logic for maintainable and scalable navigation control.

3.2.4 Risk Mitigation Plan with Alternative Solution Identification

Risk	Potential Impact	Mitigation Strategy	Alternative Solution
Unstable startup during power-on	Incorrect sensor readings or system hang	Startup delay to allow power and sensors to stabilize	External power supervisor or reset IC
ESP32 boot failure due to GPIO conflicts	System fails to boot or enters flash mode	Use of boot-safe GPIOs for sensors and motors	Remap GPIOs or use GPIO expanders
Motors activating at boot	Unintended robot movement, safety risk	Motors disabled during initialization	Hardware relay or motor power cut-off switch
Undefined navigation behavior	Erratic movement or logic failure	FSM-based control with defined states	Watchdog timer or state validation checks

3.2.5 Cost–Benefit Analysis

The system utilizes low-cost and widely available hardware components, such as IR sensors, ultrasonic modules, and the ESP32 microcontroller, reducing overall development and deployment cost.

Despite the low cost, the integration of multiple sensors and FSM-based control provides high functional reliability, improving navigation accuracy and operational safety.

The modular and scalable design allows the system to be adapted for warehouse automation applications, enabling future expansion without significant redesign or cost increase.

3.3 Performance Requirements

The system shall support real-time sensor polling to continuously monitor line position, obstacle distance, and RFID tag detection during navigation.

The motor control subsystem shall ensure an immediate motor stop upon obstacle detection, prioritizing operational safety and preventing collisions.

The system shall maintain stable PWM-based speed control, ensuring smooth and consistent motor operation during forward and reverse movement.

3.4 Design Constraints

The current prototype supports line-based navigation only, requiring a predefined black-line path for robot movement.

The system operates under limited computational and memory resources of the ESP32, requiring efficient and lightweight control logic.

3.5 Software System Attributes

3.5.1 Reliability

The reliability of the system is achieved through a multi-sensor navigation approach combined with structured control logic. The IR line sensor array provides continuous path tracking, while RFID-based location identification offers absolute positional reference points along the route. Ultrasonic sensing adds a safety layer by detecting obstacles in real time.

A finite state machine (FSM) governs all navigation decisions, ensuring that the robot transitions only between predefined states. This eliminates undefined or unexpected behaviors, particularly during critical events such as RFID detection or obstacle encounters. In the event of sensor anomalies, the FSM enforces safe actions such as stopping the motors, thereby increasing overall system robustness.

3.5.2 Availability

The system is designed to operate autonomously once powered on, without requiring manual control or user intervention during normal operation. After completing initialization and sensor stabilization, the robot continuously performs navigation, location identification, and obstacle monitoring.

Availability is further enhanced by the use of lightweight control logic, which minimizes processor load on the ESP32 and prevents system freezes or delays. This allows the robot to

remain operational for extended periods within controlled indoor environments, provided that power is continuously supplied.

3.5.3 Security

At the current prototype stage, security mechanisms are not implemented, as the system operates in an isolated and offline environment. There is no external communication, remote access, or data exchange with external systems.

Since the primary focus of this phase is hardware integration and autonomous navigation validation, security considerations such as authentication or encrypted communication are intentionally excluded and reserved for future system expansions.

3.5.4 Maintainability

The software architecture follows a modular design approach, with separate functions handling sensor input processing, motor control, and decision logic. This structure simplifies debugging, testing, and future code modifications.

The FSM-based navigation framework further improves maintainability by clearly defining system behavior through states and transitions. New behaviors, such as enhanced recovery strategies or additional sensor inputs, can be incorporated by adding new states or transitions without restructuring the existing codebase. This approach supports systematic development and long-term maintainability of the system.

3.6 Other Requirements

The system shall operate autonomously without external user interaction, performing navigation, location identification, and obstacle response automatically after initialization, with no requirement for manual control during normal operation.

All control logic shall execute locally on the ESP32 to ensure real-time response, minimizing latency by processing sensor data and motor commands directly on the embedded controller without reliance on external computation or communication.

The design shall remain modular to support future enhancements without altering the core navigation logic, enabling additional features such as advanced obstacle avoidance or recovery mechanisms to be integrated through new modules or state extensions.

4. REFERENCES

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5. APPENDICES

Appendix A : Hardware pin mapping

This appendix documents the ESP32 pin assignments used for motor control, sensor interfacing, RFID communication, and power distribution.

1. Motor Driver Connections (L298N)

Component	ESP32 Pin	Purpose
L298N IN1	GPIO 14	Left motor direction
L298N IN2	GPIO 27	Left motor direction
L298N IN3	GPIO 25	Right motor direction
L298N IN4	GPIO 26	Right motor direction
L298N ENA	GPIO 23	Left motor PWM speed
L298N ENB	GPIO 19	Right motor PWM speed

2.Sensor Connections

IR Line Sensor Array

IR Sensor	ESP32 Pin	Purpose
IR Sensor S1	GPIO 36	Leftmost line detection
IR Sensor S2	GPIO 39	Left line detection
IR Sensor S3	GPIO 34	Center (primary) sensor
IR Sensor S4	GPIO 35	Right line detection
IR Sensor S5	GPIO 13	Rightmost line detection

Ultrasonic Sensor

Ultrasonic Pin	ESP32 Pin	Purpose
TRIG	GPIO 18	Distance trigger
ECHO	GPIO 5	Distance echo

PN532 RFID Module

PN532 Pin	ESP32 Pin	Purpose
IRQ	GPIO 32	RFID interrupt signal
RESET	GPIO 33	RFID module reset
SDA	GPIO 21	I2C data
SCL	GPIO 22	I2C clock

3. Power Supply Connections

Component	Supply Voltage	Notes
ESP32	USB/5V Regulator	On-board regulation
PN532 RFID	3.3V	Logic-level compatible
IR Sensor Array	5V	Powered via 5V converter
Ultrasonic Sensor	5V	Requires stable 5V
L298N Motor Driver	12V (motor), 5V (logic)	External motor supply
Common Ground	GND	All components share common ground

Appendix B : Test Scenarios

This appendix summarizes functional test scenarios used to validate system behavior.

Test ID	Test Description	Input / Condition	Expected Result	Actual Result	Status
TC1	Line following	Robot placed on black line	Robot follows line smoothly	As expected	Pass
TC2	Line correction	Robot slightly offset from line	Robot realigns to center	As expected	Pass
TC3	RFID target detection	Target RFID tag detected	Robot stops and waits	As expected	Pass
TC4	Reverse navigation initiation	RFID target processed	Robot reverses direction	As expected	Pass
TC5	Destination RFID detection	Destination tag detected	Robot stops permanently	As expected	Pass
TC6	Obstacle detection	Obstacle within threshold distance	Motors stop immediately	As expected	Pass
TC7	Obstacle clearance	Obstacle removed	Robot resumes operation	As expected	Pass
TC8	Power-on safety	System powered ON	Motors remain OFF initially	As expected	Pass
TC9	Sensor stabilization delay	Startup delay active	No false sensor triggers	As expected	Pass

TC10	State transition validation	Normal navigation sequence	FSM transitions correctly	As expected	Pass
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